



FEU INSTITUTE OF TECHNOLOGY

COLLEGE OF COMPUTER STUDIES AND MULTIMEDIA ARTS

IT0049 (WEB SYSTEM TECHNOLOGIES)

TECHNICAL SUMMATIVE ASSESSMENT

1

TASKS FOR TODAY MANAGEMENT SYSTEM

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PROGRAM OUTCOME/S (PO) ADDRESSED BY THE LABORATORY EXERCISE

- Design, implement and evaluate computer-based systems or applications to meet desired needs and requirements. [PO: C]

COURSE LEARNING OUTCOME/S (CLO) ADDRESSED BY THE LABORATORY EXERCISE

- Apply advanced web development principles in creating, debugging, validating, and securing database-backed web applications. [CLO: 2]
- Design program flow and application structure for computing problems using CodeIgniter and Laravel frameworks. [CLO: 3]

INTENDED LEARNING OUTCOME/S (ILO) OF THE LABORATORY EXERCISE

At the end of this exercise, students must be able to:

- Design and implement a relational database schema for a small task-management application.
- Build Models and controllers that retrieve and filter records, including filtering by date.
- Produce multiple distinct views — a filtered dashboard, a full listing, a profile page, and a static page — from a shared data layer.

BACKGROUND INFORMATION

This assessment combines the two skills covered so far: organizing an application with MVC (Module 1) and backing it with a real database (Module 2). You will build a small internal tool, the Tasks for Today Management System, that a team could use to track daily to-do items.

A common pattern in dashboards like this is separating a filtered "today" view from a complete, unfiltered listing. Both views read from the same table; only the query differs. This activity asks you to implement both.



GRADING SYSTEM / RUBRIC

CRITERIA	Description
Functionality & Requirements Completeness (35 points max)	All required pages, records, and workflows are present and behave as specified in the activity instructions.
Code Structure & Organization (20 points max)	The application follows the framework's MVC conventions with a clear, consistent separation of concerns across controllers, models, and views.
Database Design & Data Retrieval (25 points max)	The schema is implemented as specified, sample data is realistic and sufficient, and queries correctly filter by date where required and retrieve all records where required.
Documentation & Submission Quality (20 points max)	The GitHub repository is organized and complete, the README explains how to set up and run the project, and the hosted link is live and matches the submitted code.



LABORATORY ACTIVITY

Build a Tasks for Today Management System with four pages: a Welcome page showing only today's tasks, a full Task List page showing every task, a Profile page showing one demo user's information, and an About page identifying you as the developer.

1. Create the database and tables using the schema below:

```
CREATE TABLE tasks (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(150) NOT NULL,  
  status VARCHAR(20) NOT NULL DEFAULT 'pending',  
  task_date DATE NOT NULL,  
  created_at DATETIME NOT NULL  
);
```

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  username VARCHAR(50) NOT NULL UNIQUE,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(100) NOT NULL,  
  created_at DATETIME NOT NULL  
);
```

2. Populate the tasks table with at least 8 records spanning at least 3 different dates, including today's date.
3. Populate the users table with exactly one demo record.
4. Build a TaskModel and a UserModel.
5. Build the Welcome page (/) to query and display only tasks where task_date equals today's date.
6. Build the Task List page (/tasks) to query and display every task, ordered by date.
7. Build the Profile page (/profile) to display the single user record.
8. Build a static About page (/about) identifying you as the developer of the system.



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Submission Requirements:

- A link to a GitHub repository containing your raw project files, including your database export or migration and seeder files.
- A link to your hosted, working version of the application.

REFERENCES

- Manfield, A. (2021). Pro CodeIgniter: Learn how to create professional web-applications with PHP.
- CodeIgniter User Guide. Query Builder — Where Clauses.
https://codeigniter4.github.io/userguide/database/query_builder.html



```
.env × Routes.php M Pages.php M env
pos-system > .env
1  # Local development settings. This file is intentionally ignored by Git.
2  CI_ENVIRONMENT = development
3
4  app.baseURL = 'http://localhost:8080/'
5  app.indexPage = ''
6
7  # Local MySQL connection for TSA1. Change these values if your setup differs.
8  database.default.hostname = 127.0.0.1
9  database.default.database = tasks_today_db
10 database.default.username = root
11 database.default.password =
12 database.default.DBDriver = MySQLi
13 database.default.DBPrefix =
14 database.default.port = 3306
15
```

Figure 1. CodeIgniter database configuration for the Tasks for Today Management System

```
[mysql> USE tasks_today_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
```

```
[mysql> SHOW TABLES;
+-----+
| Tables_in_tasks_today_db |
+-----+
| tasks                     |
| users                     |
+-----+
2 rows in set (0.006 sec)

mysql>
```

Figure 2. TSA1 MySQL database containing the required **tasks** and **users** tables



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```
[mysql> DESCRIBE tasks;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
title	varchar(150)	NO		NULL	
status	varchar(20)	NO		pending	
task_date	date	NO		NULL	
created_at	datetime	NO		NULL	

```
5 rows in set (0.011 sec)
```

```
mysql> █
```

Figure 3. Structure of the `tasks` MySQL table

```
[mysql> SELECT * FROM tasks ORDER BY task_date, id;
```

id	title	status	task_date	created_at
1	Review Web Systems lecture notes	completed	2026-10-01	2026-09-30 09:00:00
2	Organize CodeIgniter reference materials	completed	2026-10-01	2026-09-30 10:00:00
3	Complete the database integration exercise	in_progress	2026-10-02	2026-10-02 07:00:00
4	Attend the Web System Technologies lecture	pending	2026-10-02	2026-10-02 08:00:00
5	Review the TSA1 requirements	completed	2026-10-02	2026-10-02 09:00:00
6	Test the Tasks for Today pages	pending	2026-10-02	2026-10-02 10:00:00
7	Prepare notes for the next module	pending	2026-10-03	2026-10-02 11:00:00
8	Read the next CodeIgniter lesson	pending	2026-10-03	2026-10-02 12:00:00
9	Plan the next set of weekly priorities	pending	2026-10-03	2026-10-02 13:00:00

```
9 rows in set (0.002 sec)
```

```
mysql> █
```

Figure 4. Task records stored in the `tasks` table



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```
mysql> SELECT DISTINCT task_date
-> FROM tasks
-> ORDER BY task_date;
+-----+
| task_date |
+-----+
| 2026-10-01 |
| 2026-10-02 |
| 2026-10-03 |
+-----+
3 rows in set (0.003 sec)

mysql> █
```

Figure 5. Task records spanning at least three different dates

```
[mysql> DESCRIBE users;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| id         | int           | NO   | PRI | NULL    | auto_increment |
| username   | varchar(50)   | NO   | UNI | NULL    |                |
| full_name  | varchar(100)  | NO   |     | NULL    |                |
| email      | varchar(100)  | NO   |     | NULL    |                |
| created_at | datetime      | NO   |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.005 sec)

mysql> █
```

Figure 6. Structure of the `users` MySQL table



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```
[mysql> SELECT * FROM users;
+-----+-----+-----+-----+
| id | username          | full_name                | email                      | created_at |
+-----+-----+-----+-----+
| 1  | isaiahvicencio    | Isaiah Ezekiel P. Vicencio | isaiah.vicencio@example.com | 2026-10-02 08:00:00 |
+-----+-----+-----+-----+
1 row in set (0.004 sec)

mysql>
```

Figure 7. Single demo record stored in the `users` table

```
.env TaskModel.php U X Routes.php M Pages.php M
pos-system > app > Models > TaskModel.php > ...
1 | <?php
2 |
3 | namespace App\Models;
4 |
5 | use CodeIgniter\Model;
6 |
7 | 3 references | 0 implementations
8 | class TaskModel extends Model
9 | {
10 |     0 references
11 |     protected $table = 'tasks';
12 |     0 references
13 |     protected $primaryKey = 'id';
14 |     0 references
15 |     protected $returnType = 'array';
16 |
17 |     0 references
18 |     protected $allowedFields = [
19 |         'title',
20 |         'status',
21 |         'task_date',
22 |         'created_at',
23 |     ];
24 |
25 |     1 reference | 0 overrides
26 |     public function findForDate(string $date): array
27 |     {
28 |         return $this->where('task_date', $date)
29 |             ->orderBy('created_at', 'ASC')
30 |             ->findAll();
31 |     }
32 |
33 |     1 reference | 0 overrides
34 |     public function findAllOrdered(): array
35 |     {
36 |         return $this->orderBy('task_date', 'ASC')
37 |             ->orderBy('id', 'ASC')
38 |             ->findAll();
39 |     }
40 | }
```

Figure 8. TaskModel configured for the `tasks` database table



```
.env UserModel.php M Routes.php M Pages.php M
pos-system > app > Models > UserModel.php > ...
1  <?php
2
3  namespace App\Models;
4
5  use CodeIgniter\Model;
6
7  2 references | 0 implementations
8  class UserModel extends Model
9  {
10     0 references
11     protected $table = 'users';
12     0 references
13     protected $primaryKey = 'id';
14     0 references
15     protected $returnType = 'array';
16
17     0 references
18     protected $allowedFields = [
19         'username',
20         'full_name',
21         'email',
22         'created_at',
23     ];
24 }
```

Figure 9. UserModel configured for the **users** database table

Tasks for Today

Daily task manager

WORKSPACE

Today

All Tasks

Profile

About

Task Management / Today's Tasks

CodeIgniter 4 - MySQL

DAILY FOCUS

Tasks for today

4 DUE TODAY

Friday, October 2, 2026 – only tasks scheduled for today are shown here.

TODAY'S SCHEDULE

Current tasks

tasks table

ID	TASK	STATUS	TASK DATE	CREATED AT
#3	Complete the database integration exercise	In Progress	Oct 2, 2026	Oct 2, 2026 - 7:00 AM
#4	Attend the Web System Technologies lecture	Pending	Oct 2, 2026	Oct 2, 2026 - 8:00 AM
#5	Review the TSA1 requirements	Completed	Oct 2, 2026	Oct 2, 2026 - 9:00 AM
#6	Test the Tasks for Today pages	Pending	Oct 2, 2026	Oct 2, 2026 - 10:00 AM

View all tasks

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Technical Summative Assessment 1

Tasks for Today - Technical Summative Assessment 1

IT0049 - Web System Technologies

Figure 10. Welcome page displaying only today's tasks



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Task Management / All Tasks

CodeIgniter 4 - MySQL

Tasks for Today
Daily task manager

WORKSPACE

- Today
- All Tasks
- Profile
- About

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COMPLETE SCHEDULE

All tasks

Every task in the database, ordered by its scheduled date.

TOTAL TASKS: 9

TASK REGISTER

All scheduled tasks

ordered by task_date

ID	TASK	STATUS	TASK DATE	CREATED AT
#1	Review Web Systems lecture notes	Completed	Oct 1, 2026	Sep 30, 2026 · 9:00 AM
#2	Organize CodeIgniter reference materials	Completed	Oct 1, 2026	Sep 30, 2026 · 10:00 AM
#3	Complete the database integration exercise	In Progress	Oct 2, 2026	Oct 2, 2026 · 7:00 AM
#4	Attend the Web System Technologies lecture	Pending	Oct 2, 2026	Oct 2, 2026 · 8:00 AM
#5	Review the TSA1 requirements	Completed	Oct 2, 2026	Oct 2, 2026 · 9:00 AM
#6	Test the Tasks for Today pages	Pending	Oct 2, 2026	Oct 2, 2026 · 10:00 AM
#7	Prepare notes for the next module	Pending	Oct 3, 2026	Oct 2, 2026 · 11:00 AM
#8	Read the next CodeIgniter lesson	Pending	Oct 3, 2026	Oct 2, 2026 · 12:00 PM
#9	Plan the next set of weekly priorities	Pending	Oct 3, 2026	Oct 2, 2026 · 1:00 PM

Figure 11. Full Task List displaying every task ordered by date

Task Management / Profile

CodeIgniter 4 - MySQL

Tasks for Today
Daily task manager

WORKSPACE

- Today
- All Tasks
- Profile
- About

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ACCOUNT PROFILE

Developer profile

The single demonstration user stored in the application database.

DATABASE USER

Isaiah Ezekiel P. Vicencio
@isaiahvicencio

Username: isaiahvicencio

Full name: Isaiah Ezekiel P. Vicencio

Email: isaiah.vicencio@example.com

Created: October 2, 2026 · 8:00 AM

Figure 12. Profile page displaying the single demo user record



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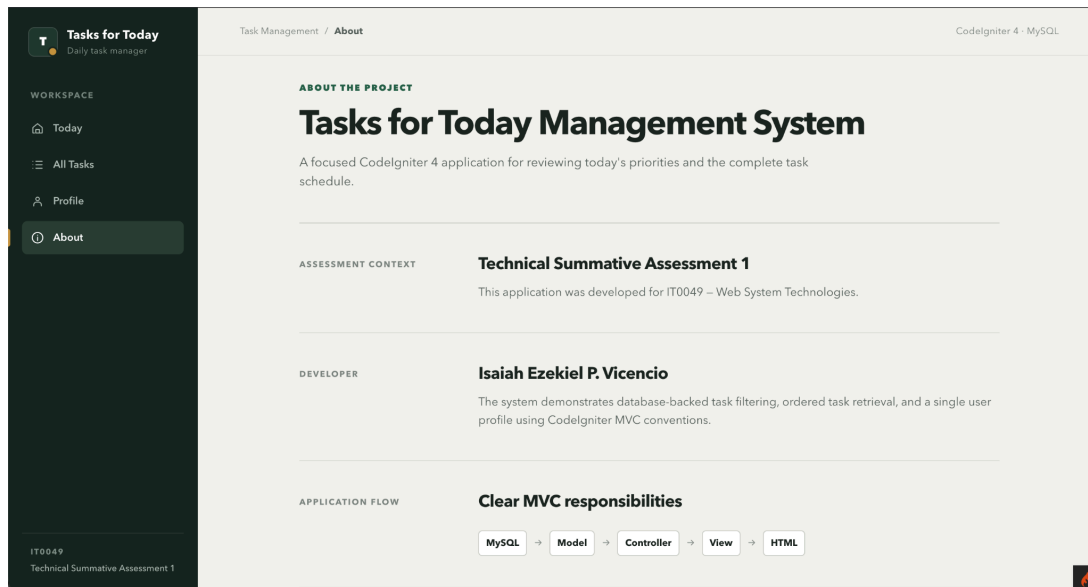


Figure 13. Static About page identifying the system developer

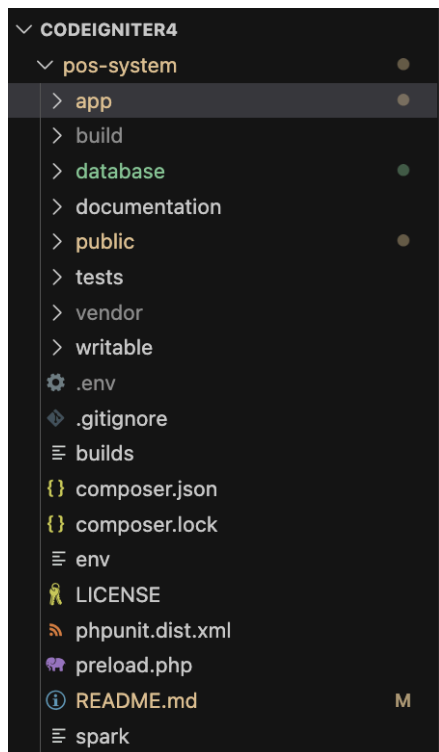


Figure 14. CodeIgniter MVC project structure